Assurance of Learning

COMP6799049 - Database Technology

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“Database Financial Technology: GOPAY”



LA06

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Chapter 1

Identification of Performance Problem

A. Case Description

In today's fast-paced digital era, financial technology reshapes how people manage and move their money. One major player in Indonesia's fintech industry is GoPay, a digital wallet that enables users to make secure, cashless transactions with ease. Users can maintain a personal wallet containing both a real balance and coin balance. This setup supports everyday spending as well as promotional rewards.

GoPay supports various transaction types, including top-ups, transfers, merchant payments, cashbacks, and refunds. These transactions occur seamlessly within a unified ecosystem, making the platform flexible and user-friendly. To support personalization and operational accuracy, each user and merchant is linked to structured location data. This information helps enhance services such as regional promotions and fraud detection.

Given the complexity and volume of transactions, a robust database structure is essential. Raw, unnormalized data often contains redundancy and inconsistencies that can harm performance. Therefore, a refined, normalized database is needed to maintain integrity and support scalability. This project focuses on identifying anomalies and redesigning the schema using normalization techniques.

The final goal is to develop a clean and efficient Entity-Relationship Diagram (ERD) that accurately models GoPay's operations. This design reflects core principles of relational databases while addressing real-world fintech challenges. Through this case study, theoretical concepts are applied to a practical system, showing how proper data modeling supports reliable digital financial services.

Chapter 2

Preparation for the sample data

A. Sampel Data

user_details	wallet_info	transaction_info	merchant_details	location_info	payment_details
USR001, John, +6281234567890, john.doe@email.com, 1990-05-15, debit 1234, true	WLT001, 500000, 100	TRX001, payment, 75000 IDR, completed, 2025-01-05 13:45:22	MERCH001, Coffee Shop A, GOPAY, 8723456789	LOC001, Jl. Sudirman 123, Jakarta, DKI Jakarta, 12190, Indonesia	null, ORD123, null, null, null, null
USR002, Jane, +6287654321098, jane.smith@email.com, 1995-11-22, debit, 5678, true	WLT002, 750000, 250	TRX002, top_up, 250000 IDR, completed, 2025-01-07 09:12:34	null	LOC002, Jl. Thamrin 45, Jakarta, DKI Jakarta, 10350, Indonesia	debit_card, null, null, null, null, null
USR003, Mike, +6289876543210, mike.brown@email.com, 1988-07-30, merchant, 9012, true	WLT003, 1250000, 500	TRX003, money_transfer, 100000 IDR, completed, 2025-01-08 16:30:45	null	LOC003, Jl. Asia Afrika 78, Bandung, West Java, 40112, Indonesia	null, null, TRX001, null, null, null
USR005, Johnny, +6281234567891, johnny@email.com, 1990-05-15, merchant, 1324, true	WLT005, 550000, 150	TRX004, withdrawal, 200000 IDR, completed, 2025-01-10 11:20:15	null	LOC005, Jl. Sudirman 124, Jakarta, DKI Jakarta, 12191, Indonesia	null, null, null, null, null, BCA, null, null
USR004, Sara +6282345678901, sara.lee@email.com, 1992-03-18, bank, 3456, true	WLT004, 900000, 150	TRX005, payment, 150000 IDR, completed, 2025-01-12 19:05:33	MERCH002, Grocery Store, GOPAY, 9876543210, Surabaya	LOC004, Jl. Merdeka 56, Surabaya, East Java, 60231, Indonesia	null, ORD456, null, null, null, null

USR006, Janet, +6287654321099, janet@email.com, 1995-11-22, 5768, true	WLT006, 700000, 230	TRX007, refund, 50000 IDR, completed, 2025-01-15 14:40:27	MERCH003, Coffee Shop B, GOPAY, 8723456788	LOC006, Jl. Thamrin 46, Jakarta, DKI Jakarta, 10351, Indonesia	null, null, null, null, null, TRX006, null

B. Sample Data Description

Variable	Description
user_details	Contains complete user information in the form of ID, name, telephone number, email, date of birth, PIN, and account activeness status.
wallet_info	Displays the wallet ID, available balance, and bonus amount owned by the user.
transaction_info	Describes transaction details including ID, transaction type, amount of money, status, and transaction time
merchant_details	Contains merchant information such as ID, store name, payment platform used, contact number, and location.
location_info	Provide the full address where the transaction is taking place, including street, city, province, postal code, and country.
payment_details	It contains the payment method, order ID, as well as additional information such as the user or related transaction and bank name if any.

Chapter 3

Overcoming and Recognizing Data Anomalies and Functional Dependencies

A. List of anomalies that may occur

Based on the analysis of the GoPay data structure that has not been normalized, the following are anomalies that can occur in the database system:

1) Redundancy Problem (Data Redundancy)

- Description:

  User and merchant address data will repeat if there are different transactions involving the same entity.

  Case Example:

  1) When user USR001 (John) performs multiple transactions from the same location (LOC001 - Jl. Sudirman 123, Jakarta), the address information will be stored repeatedly in each transaction record
  2) Merchant MERCH001 (Coffee Shop A) that serves multiple transactions will store the same information repeatedly.

- Impact:

  a) Waste of storage space
  b) Inconsistency of data if there is a change of address
  c) Maintenance is difficult and error-prone

- Solution:

  Separation of Location table as a separate entity with foreign key reference

2) Insertion Anomaly

- Description:

Unable to add a new merchant without a transaction first.

Case Example:

1) The system cannot register a new merchant such as "Restaurant XYZ" unless a transaction occurs
2) New user data cannot be inserted into the system without a wallet and transaction.

- Impact:
    a) Limitations in master data management
    b) Inflexible for new merchant/user registration
    c) Forced creation of dummy data for registration

- Solution:

    Separation of User, Merchant, and Transaction entities into separate tables

3) Update Anomaly

- Description:

    Changing merchant address or user information must be done on multiple lines simultaneously.

    Case Example:

1) If Coffee Shop A (MERCH001) moves its address from "Jl. Sudirman 123" to a new address, then all transaction records involving the merchant must be updated
2) Changes to a user's phone number must be made in all transaction records involving that user.

- Impact:
    a) Complex and time-consuming update process
    b) Risk of inconsistency if not all records are updated
    c) Performance degradation on update operations

- Solution:

  Normalize with foreign key relationships so that updates are only done in one place

4) Deletion Anomaly

- Description:

  If a transaction is deleted, the user or merchant data can be lost.

  Case Example:

1) If transaction TRX001 is deleted, information about user USR001 (John) and merchant MERCH001 (Coffee Shop A) will also disappear from the system
2) Deleting the last transaction of a merchant will remove the merchant data from the database.

- Impact:
  a) Loss of critical business data
  b) Inability to maintain historical data
  c) Difficulty in audit and compliance

- Solution:

  Separation of entities with proper referential integrity constraints

5) Data Consistency Anomalies

- Description:

  Inconsistencies in data formatting and validation that can lead to operational errors.

  Case Example:

1) Inconsistent phone number formats (some with +62, some without)
2) Non-standard or missing currency codes
3) Non-uniform transaction statuses

- Impact:
    a) Data quality issues
    b) Difficulties in reporting and analytics
    c) Errors in business logic processing

- Solution:
    Implementation of proper data types, constraints, and validation rules

6) Functional Dependency Anomalies

- Description:
    Improper functional dependencies can cause anomalies in database operations.

Examples of problematic Functional Dependencies:

1) transaction_id → user_id, merchant_id, location_id, payment_method
2) user_id → full_name, phone_number, email, date_of_birth, pin
3) merchant_id → merchant_name, acquirer_name, merchant_pan

- Impact:
    a) Partial dependencies cause redundancy
    b) Transitive dependencies cause update anomalies
    c) Complex composite key dependencies

- Solution:
    Apply normalization to 3NF to remove partial and transitive dependencies

Chapter 4

Implementing Normal Form Techniques

A. Normalization Implementation

1) Normalization U-NF to 1-NF

1-NF Table:

user_id (PK)	full_name	phone_number	email	date_of_birth	pin	status_user	wallet_id (PK)	balance	coin_balance	transaction_id (FK)	transaction_type	amount	currency_code	status	created_at	time_hour	topup_method	order_id	reciever_user_id	withdrawal_bank	original_transaction_id	cashback_source	merchant_id (FK)	merchant_name	acquirer_name	merchant_pan	location_id (PK)	street	city	province	postal_code	Country
USR001	John	6281234567890	john.doe@email.com	5/15/1990	1234	TRUE	WLT001	5000000	100	TRX001	payment	75000	IDR	completed	1/5/2025	13:45:22	-	ORD123	-	-	-	-	-	Coffee Shop A	GOPAY	8723456789	LOC001	Jl. Sudirman 123	Jakarta	DKI Jakarta	12190	Indonesia
USR002	Jane	6287654321098	jane.smith@email.com	11/22/1995	5678	TRUE	WLT002	750000	250	TRX002	top_up	250000	IDR	completed	1/7/2025	9:12:34 AM	debit_card	-	-	-	-	-	-	-	-	LOC002	Jl. Thamrin 45	Jakarta	DKI Jakarta	10350	Indonesia
USR003	Mike	6289876543210	mike.brown@email.com	7/30/1988	9012	TRUE	WLT003	1250000	500	TRX003	money_transfer	100000	IDR	completed	1/8/2025	16:30:45	-	-	TRX001	-	-	-	-	-	-	-	LOC003	Jl. Asia Afrika 78	Jakarta	DKI Jakarta	40112	Indonesia
USR004	Sara	6282345678901	sara.lee@email.com	3/18/1992	3456	TRUE	WLT004	900000	150	TRX004	payment	150000	IDR	completed	1/12/2025	19:05:33	-	ORD456	-	-	-	-	MERCH002	Grocery Store	GOPAY	9876543210	LOC004	Jl. Kebon Jeruk 56	Jakarta	DKI Jakarta	60231	Indonesia
USR005	Johnny	6282345678901	johnny@email.com	5/15/1990	1324	TRUE	WLT005	350000	150	TRX005	withdrawal	200000	IDR	completed	1/10/2025	11:20:15	-	-	-	BCA	-	-	-	-	-	-	LOC005	Jl. Sudirman 124	Jakarta	DKI Jakarta	12190	Indonesia
USR006	Janet	6287654321099	janet@email.com	11/22/1995	5768	TRUE	WLT006	700000	230	TRX007	refund	50000	IDR	completed	1/15/2025	14:40:27	-	-	-	-	TRX006	-	MERCH003	Coffee Shop B	GOPAY	8723456789	LOC006	Jl. Thamrin 46	Jakarta	DKI Jakarta	10351	Indonesia

At this stage, data that is still in Unnormalized Form (UNF) is converted into 1NF form by:

- Breaking data that contains repeating groups or multivalued values.
- Ensuring that each column only contains atomic values (single and indivisible).
- Splitting the combined data into different entities

➢ Normalization Details, **each table in 1NF form still in one single table**:

1. User_Details

user_details
USR001, John, +6281234567890, john.doe@email.com, 1990-05-15, debit 1234, true
USR002, Jane, +6287654321098, jane.smith@email.com, 1995-11-22, debit, 5678, true
USR003, Mike, +6289876543210, mike.brown@email.com, 1988-07-30, merchant, 9012, true
USR005, Johnny, +6281234567891, johnny@email.com, 1990-05-15, merchant, 1324, true
USR004, Sara +6282345678901, sara.lee@email.com, 1992-03-18, bank, 3456, true
USR006, Janet, +6287654321099, janet@email.com, 1995-11-22, 5768, true

After normalization by making data in the database has atomic value in every variable:

user_id (PK)	full_name	phone_number	email	date_of_birth	pin	status_user
USR001	John	6281234567890	john.doe@email.com	5/15/1990	1234	TRUE
USR002	Jane	6287654321098	jane.smith@email.com	11/22/1995	5678	TRUE
USR003	Mike	6289876543210	mike.brown@email.com	7/30/1988	9012	TRUE
USR004	Sara	6282345678901	sara.lee@email.com	3/18/1992	3456	TRUE
USR005	Johnny	6282345678901	johnny@email.com	5/15/1990	1324	TRUE
USR006	Janet	6287654321099	janet@email.com	11/22/1995	5768	TRUE

2. Wallet_Info

wallet_info
WLT001, 500000, 100
WLT002, 750000, 250
WLT003, 1250000, 500
WLT005, 550000, 150
WLT004, 900000, 150
WLT006, 700000, 230

After normalization by making data in the database has atomic value in every variable:

wallet_id (PK)	balance	coin_balance
WLT001	5000000	100
WLT002	750000	250
WLT003	1250000	500
WLT004	900000	150
WLT005	550000	150
WLT006	700000	230

3. Transaction_Info

transaction_info
TRX001, payment, 75000 IDR, completed, 2025-01-05 13:45:22
TRX002, top_up, 250000 IDR, completed, 2025-01-07 09:12:34
TRX003, money_transfer, 100000 IDR, completed, 2025-01-08 16:30:45
TRX004, withdrawal, 200000 IDR, completed, 2025-01-10 11:20:15
TRX005, payment, 150000 IDR, completed, 2025-01-12 19:05:33
TRX007, refund, 50000 IDR, completed, 2025-01-15 14:40:27

After normalization by making data in the database has atomic value in every variable:

transaction_id (PK)	transaction_type	amount	currency_code	status	created_at	time_hour	topup_method	order_id	reciever_user_id	withdrawal_bank	original_transaction_id	cashback_source
TRX001	payment	75000	IDR	completed	1/5/2025	13:45:22	-	ORD123	-	-	-	-
TRX002	top_up	250000	IDR	completed	1/7/2025	9:12:34 AM	debit_card	-	-	-	-	-
TRX003	money_transfer	100000	IDR	completed	1/8/2025	16:30:45	-	-	TRX001	-	-	-
TRX004	payment	150000	IDR	completed	1/12/2025	19:05:33	-	ORD456	-	-	-	-
TRX005	withdrawal	200000	IDR	completed	1/10/2025	11:20:15	-	-	-	BCA	-	-
TRX007	refund	50000	IDR	completed	1/15/2025	14:40:27	-	-	-	-	TRX006	-

4. Merchant_Details

merchant_details
MERCH001, Coffee Shop A, GOPAY, 8723456789

null
null
null
MERCH002, Grocery Store, GOPAY, 9876543210, Surabaya
MERCH003, Coffee Shop B, GOPAY, 8723456788

After normalization by making data in the database has atomic value in every variable:

merchant_id(PK)	merchant_name	acquirer_name	merchant_pan
MERCH001	Coffee Shop A	GOPAY	8723456789
-	-	-	-
-	-	-	-
MERCH002	Grocery Store	GOPAY	9876543210
-	-	-	-
MERCH003	Coffee Shop B	GOPAY	8723456788

5. Location_Info

location_info
LOC001, Jl. Sudirman 123, Jakarta, DKI Jakarta, 12190, Indonesia
LOC002, Jl. Thamrin 45, Jakarta, DKI Jakarta, 10350, Indonesia
LOC003, Jl. Asia Afrika 78, Bandung, West Java, 40112, Indonesia
LOC005, Jl. Sudirman 124, Jakarta, DKI Jakarta, 12191, Indonesia
LOC004, Jl. Merdeka 56, Surabaya, East Java, 60231, Indonesia
LOC006, Jl. Thamrin 46, Jakarta, DKI Jakarta, 10351, Indonesia

After normalization by making data in the database has atomic value in every variable:

location_id(PK)	street	city	province	postal_code	Country
LOC001	Jl. Sudirman 123	Jakarta	DKI Jakarta	12190	Indonesia
LOC002	Jl. Thamrin 45	Jakarta	DKI Jakarta	10350	Indonesia

LOC003	Jl. Asia Afrika 78	Jakarta	DKI Jakarta	40112	Indonesia
LOC004	Jl. Merdeka 56	Jakarta	DKI Jakarta	60231	Indonesia
LOC005	Jl. Sudirman 124	Jakarta	DKI Jakarta	12191	Indonesia
LOC006	Jl. Thamrin 46	Jakarta	DKI Jakarta	10351	Indonesia

6. Payment_Details

payment_details
null, ORD123, null, null, null, null
debit_card, null, null, null, null, null
null, null, TRX001, null, null, null
null, null, null, null, BCA, null, null
null, ORD456, null, null, null, null
null, null, null, null, null, TRX006, null

After 1NF normalization, made the data in atomic value:

topup_method	order_id	receiver_user_id	withdrawal_bank	original_transaction_id	cashback_source
-	ORD123	-	-	-	-
debit_card	-	-	-	-	-
-	-	TRX001	-	-	-
-	ORD456	-	-	-	-
-	-	-	BCA	-	-
-	-	-	-	TRX006	-

Discussion:

All tables that have been normalized to 1-NF are still in 1 joined table.

1.1. Functional Dependency in the table

Since it's all still joined as 1 table, this causes dependencies to appear, such as:

a) Partial Dependency

This condition applies when a table has more than one primary key, causing a non-prime attribute to only depend on a part of a composite primary key, not the whole key. For example, on the table above, it contains more than one primary key which causes partial dependencies to appear such as for example.

Example based on the table above:

1) full_name only depends on the user's id but since there's multiple primary keys, this causes the full_name to also depend on the other primary keys. So, even if the other primary keys are deleted except for the user's id, the full_name can still fully function since its primary dependency lies with the user's_id. This also applies to the other non-prime attributes.

b) Transitive Dependency

This condition applies when a non-prime attribute depends on another non-prime attribute instead of the primary key which leads to data redundancy.

Example based on the table above:

1) transaction_type determines whether the attribute topup_method, order_id, reciever_user_id, withdrawal_bank, original_transaction_id, cashback_course have values within them since a transaction_type can only fill one of the attributes everytime a new data is inserted. So instead of depending on the primary key, they depend on transaction_type which is a non-prime attribute.

2) Normalization 1-NF to 2-NF

1-NF Table:

user_id (PK)	full_name	phone_number	email	date_of_birth	pin	status_user	wallet_id (PK)	balance	coin_balance	transaction_id (PK)	transaction_type	amount	currency_code	status	created_at	time_hour	topup_method	order_id	reciever_user_id	withdrawal_bank	original_transaction_id	cashback_source	merchant_id(PK)	merchant_name	acquirer_name	merchant_pan	location_id(PK)	street	city	province	postal_code	Country
USR001	John	6281234567890	john.doe@email.com	5/15/1990	1234	TRUE	WLT001	5000000	100	TRX001	payment	75000	IDR	completed	1/5/2025	13:45:22	-	ORD123	-	-	-	-	MERCH001	Coffee Shop A	GOPAY	87234567890	LOC001	Jl. Sudirman 123	Jakarta	DKI Jakarta	12190	Indonesia
USR002	Jane	6287654321098	jane.smith@email.com	11/22/1995	5678	TRUE	WLT002	750000	250	TRX002	top_up	250000	IDR	completed	1/7/2025	9:12:34 AM	debit_card	-	-	-	-	-	-	-	-	-	LOC002	Jl. Thamrin 45	Jakarta	DKI Jakarta	10350	Indonesia
USR003	Mike	6289876543210	mike.brown@email.com	7/30/1988	9012	TRUE	WLT003	1250000	500	TRX003	money_transfer	100000	IDR	completed	1/8/2025	16:30:45	-	-	TRX001	-	-	-	-	-	-	-	LOC003	Jl. Asia Afrika 78	Jakarta	DKI Jakarta	40112	Indonesia
USR004	Sara	6282345678901	sara.lee@email.com	3/18/1992	3456	TRUE	WLT004	900000	150	TRX004	payment	150000	IDR	completed	1/12/2025	19:05:33	-	ORD456	-	-	-	-	MERCH002	Grocery Store	GOPAY	9876543210	LOC004	Jl. Juanda 56	Jakarta	DKI Jakarta	60231	Indonesia
USR005	Johnny	6282345678901	johnny@email.com	5/15/1990	1324	TRUE	WLT005	550000	150	TRX005	withdrawal	200000	IDR	completed	1/10/2025	11:20:15	-	-	-	BCA	-	-	-	-	-	-	LOC005	Jl. Sudirman 124	Jakarta	DKI Jakarta	12191	Indonesia
USR006	Janet	6287654321099	janet@email.com	11/22/1995	5768	TRUE	WLT006	700000	230	TRX007	refund	50000	IDR	completed	1/15/2025	14:40:27	-	-	-	-	TRX006	-	MERCH003	Coffee Shop B	GOPAY	87234567890	LOC006	Jl. Thamrin 46	Jakarta	DKI Jakarta	10351	Indonesia

1-NF to 2-NF aims to remove partial dependencies:

- If there is a composite key, all non-primary columns must depend on the entire key part.
- Split the attributes that only partially depend on the composite key to a new table.

➢ 2NF normalization details:

1. User_Details

user_id (PK)	full_name	phone_number	email	date_of_birth	pin	status_user
USR001	John	6281234567890	john.doe@email.com	5/15/1990	1234	TRUE
USR002	Jane	6287654321098	jane.smith@email.com	11/22/1995	5678	TRUE
USR003	Mike	6289876543210	mike.brown@email.com	7/30/1988	9012	TRUE
USR004	Sara	6282345678901	sara.lee@email.com	3/18/1992	3456	TRUE
USR005	Johnny	6282345678901	johnny@email.com	5/15/1990	1324	TRUE
USR006	Janet	6287654321099	janet@email.com	11/22/1995	5768	TRUE

2. Wallet_Info

wallet_id (PK)	balance	coin_balance
WLT001	5000000	100
WLT002	750000	250
WLT003	1250000	500
WLT004	900000	150
WLT005	550000	150

3. Transaction_Info

transaction_id (PK)	transaction_type	amount	currency_code	status	created_at	time_hour	topup_method	order_id	reciever_user_id	withdrawal_bank	original_transaction_id	cashback_source
TRX001	payment	75000	IDR	completed	1/5/2025	13:45:22	-	ORD123	-	-	-	-
TRX002	top_up	250000	IDR	completed	1/7/2025	9:12:34 AM	debit_card	-	-	-	-	-
TRX003	money_transfer	100000	IDR	completed	1/8/2025	16:30:45	-	-	TRX001	-	-	-
TRX004	payment	150000	IDR	completed	1/12/2025	19:05:33	-	ORD456	-	-	-	-
TRX005	withdrawal	200000	IDR	completed	1/10/2025	11:20:15	-	-	-	BCA	-	-
TRX007	refund	50000	IDR	completed	1/15/2025	14:40:27	-	-	-	-	TRX006	-

4. Merchant_Details

merchant_id(PK)	merchant_name	acquirer_name	merchant_pan
MERCH001	Coffee Shop A	GOPAY	8723456789
-	-	-	-
-	-	-	-
MERCH002	Grocery Store	GOPAY	9876543210
-	-	-	-
MERCH003	Coffee Shop B	GOPAY	8723456788

5. Location_Info

location_id(PK)	street	city	province	postal_code	Country
LOC001	Jl. Sudirman 123	Jakarta	DKI Jakarta	12190	Indonesia
LOC002	Jl. Thamrin 45	Jakarta	DKI Jakarta	10350	Indonesia
LOC003	Jl. Asia Afrika 78	Jakarta	DKI Jakarta	40112	Indonesia
LOC004	Jl. Merdeka 56	Jakarta	DKI Jakarta	60231	Indonesia
LOC005	Jl. Sudirman 124	Jakarta	DKI Jakarta	12191	Indonesia
LOC006	Jl. Thamrin 46	Jakarta	DKI Jakarta	10351	Indonesia

6. Payment_Details

topup_method	order_id	receiver_user_id	withdrawal_bank	original_transaction_id	cashback_source
-	ORD123	-	-	-	-
debit_card	-	-	-	-	-
-	-	TRX001	-	-	-
-	ORD456	-	-	-	-
-	-	-	BCA	-	-
-	-	-	-	TRX006	-

Discussion:

All tables that have been normalized from 1-NF to 2-NF have been separated into multiple tables. In this case, the 1-NF table has been split into 6 tables or 6 entities.

3) Normalization 2-NF to 3-NF

2NF to 3NF aims to remove transitive dependencies:

1. Remove transitive dependencies - i.e. non-primary attributes that depend on other non-primary attributes.
2. Objective: ensure all non-primary columns only depend directly on the primary key.

➢ 3NF Normalization Details:

1. User Table

user_id (PK)	full_name	phone_number	email	date_of_birth	pin	status_user
USR001	John	6281234567890	john.doe@email.com	5/15/1990	1234	TRUE
USR002	Jane	6287654321098	jane.smith@email.com	11/22/1995	5678	TRUE
USR003	Mike	6289876543210	mike.brown@email.com	7/30/1988	9012	TRUE
USR004	Sara	6282345678901	sara.lee@email.com	3/18/1992	3456	TRUE
USR005	Johnny	6282345678901	johnny@email.com	5/15/1990	1324	TRUE
USR006	Janet	6287654321099	janet@email.com	11/22/1995	5768	TRUE

2. Wallet Table

wallet_id (PK)	balance	coin_balance
WLT001	5000000	100
WLT002	750000	250
WLT003	1250000	500
WLT004	900000	150
WLT005	550000	150
WLT006	700000	230

3. Transaction Info Table

transaction_id (PK)	transaction_type	amount	currency_code	status	created_at	time_hour
TRX001	payment	75000	IDR	completed	1/5/2025	13:45:22
TRX002	top_up	250000	IDR	completed	1/7/2025	9:12:34
TRX003	money_transfer	100000	IDR	completed	1/8/2025	16:30:45
TRX004	payment	150000	IDR	completed	1/12/2025	19:05:33
TRX005	withdrawal	200000	IDR	completed	1/10/2025	11:20:15
TRX006	refund	50000	IDR	completed	1/15/2025	14:40:27

4. Transaction Payment Table

transaction_id (FK)	merchant_id(FK)	order_id
TRX001	MERCH001	ORD123
TRX004	MERCH002	ORD456
TRX003	money_transfer	100000
TRX004	payment	150000
TRX005	withdrawal	200000
TRX006	refund	50000

5. Transaction Cashback Table

transaction_id (FK)	cashback_source

6. Transaction Transfer Table

transaction_id (FK)	reciever_user_id
TRX003	TRX001

7. Transaction Withdrawal Table

transaction_id (FK)	withdrawal_bank
TRX005	BCA

8. Transaction Refund Table

transaction_id (FK)	original_transaction_id
TRX007	TRX006

9. Transaction Topup Table

transaction_id (FK)	topup_method
TRX002	debit_card

10. Location Table

location_id(PK)	street	city	province	postal_code	Country
LOC001	Jl. Sudirman 123	Jakarta	DKI Jakarta	12190	Indonesia
LOC002	Jl. Thamrin 45	Jakarta	DKI Jakarta	10350	Indonesia
LOC003	Jl. Asia Afrika 78	Jakarta	DKI Jakarta	40112	Indonesia
LOC004	Jl. Merdeka 56	Jakarta	DKI Jakarta	60231	Indonesia

LOC005	Jl. Sudirman 124	Jakarta	DKI Jakarta	12191	Indonesia
LOC006	Jl. Thamrin 46	Jakarta	DKI Jakarta	10351	Indonesia

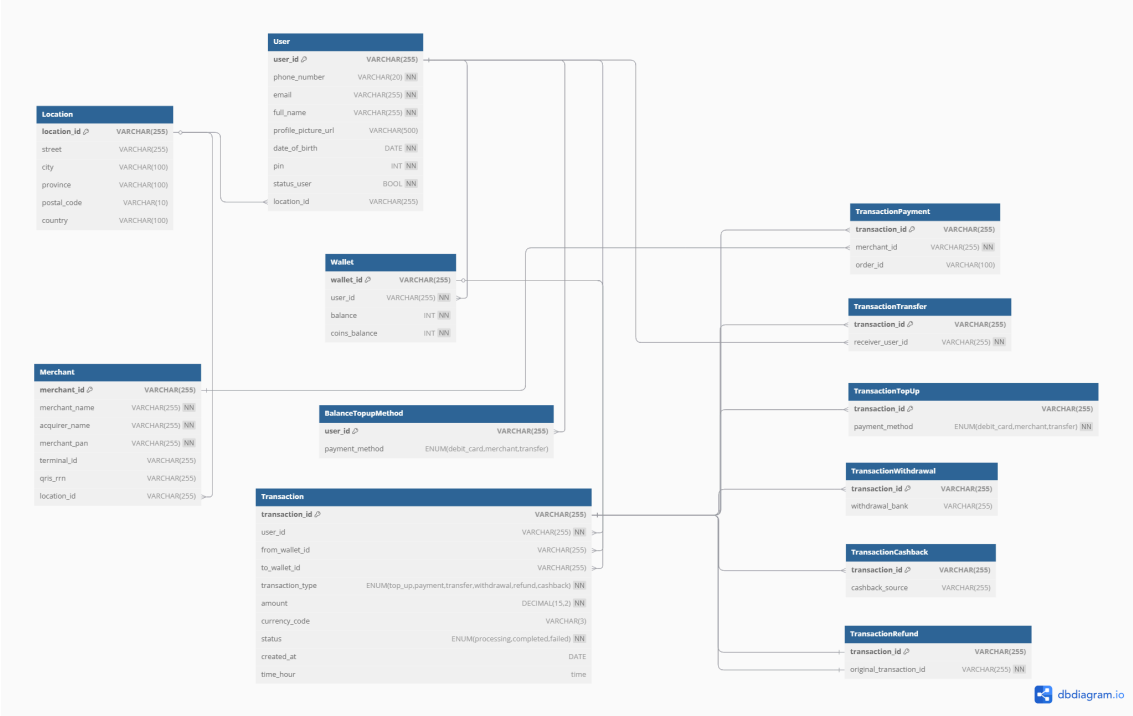
11. Merchant Table

merchant_id(PK)	**merchant_name**	**acquirer_name**	**merchant_pan**
MERCH001	Coffee Shop A	GOPAY	8723456789
-	-	-	-
-	-	-	-
MERCH002	Grocery Store	GOPAY	9876543210
-	-	-	-
MERCH003	Coffee Shop B	GOPAY	8723456788

Chapter 5

Implementing Normal Form Techniques

A. *Entity Relational Diagram*



After a structured and thorough normalization process from unnormalized form to 3NF, the final Entity-Relationship Diagram (ERD) represents a well-organized and anomaly-free relational model for the digital wallet system (GOPAY).

Initially, the system faced potential data issues such as update anomalies, redundant information, and data inconsistencies due to unstructured sample data. Through identifying functional dependencies and systematically applying normalization rules, these issues have been resolved, resulting in a robust and normalized database schema.

The final ERD consists of multiple core and supporting entities:

1. User

   Represents individual account holders. Each user is associated with personal details such as name, phone number, email, date of birth, and a security PIN. A foreign key connects the user to a Location table, allowing geographic information (such as city, province, and country) to be stored separately for better reusability and to reduce redundancy.

2. Wallet

   Each user has exactly one wallet. This table stores the balance and coins balance, helping represent the financial state of a user. It is directly linked to the User table via user_id.

3. Transaction

   This is the central entity for capturing all financial activities. It includes key attributes like the source and destination wallet (from_wallet_id, to_wallet_id), transaction type (top_up, payment, transfer, etc.), amount, currency, status, and timestamp information. The generalized structure of this table allows the system to consistently log and manage various kinds of transactions. The table is extended into specialized transaction types through the following sub-entities:

   a. TransactionTopUp

      Captures information for top-up transactions, including the method used (debit card, merchant, or transfer).

   b. TransactionPayment

      Records payments made to merchants, including merchant ID and order ID.

   c. TransactionTransfer

Logs peer-to-peer transfers between users.

d. TransactionWithdrawal

Stores withdrawal transaction data, such as the target bank.

e. TransactionCashback

Logs cashback credits given to users, including the cashback source.

f. TransactionRefund

Records refunds by linking to the original transaction.

This modular transaction model improves data consistency, clarity, and makes it easier to manage, trace, and analyze different transaction types.

4. Merchant

Represents third-party vendors or partners in the system. Each merchant has identifying information such as name, acquirer details, QRIS information, terminal ID, and location. This entity supports payment-related transactions from users to merchants.

5. BalanceTopupMethod

Captures the preferred balance top-up methods of users. It stores the user_id and their chosen method, enhancing user personalization and flexibility.

6. Location

This supporting entity stores geographical data like street, city, province, postal code, and country. By separating this information, both users and merchants can reference a common location structure, which promotes reusability and minimizes redundancy.

Each table in the refined ERD was created based on full functional dependencies and with transitive dependencies removed, ensuring that:

- Each entity focuses on a single type of data,
- No data is redundantly stored across tables,
- Relationships between entities are logically structured and enforced using foreign keys.

This relational model enables the system to:

- Efficiently track, categorize, and manage a variety of transaction types,
- Prevent data anomalies such as duplication and inconsistency,
- Scale to support advanced features in the future such as loyalty programs, user tiers, or fraud detection mechanisms.

B. Conclusion

This refined ERD is the result of a structured and methodical normalization process that prioritizes data integrity, scalability, and system performance. It demonstrates how applying normalization principles—specifically up to the Third Normal Form (3NF)—helps eliminate redundancy, resolve data anomalies, and produce a clean, modular database structure. The clear separation between user data, wallet balances, transaction types, merchant information, and geographic data allows for efficient maintenance and future scalability.

By organizing transactions into a generalized parent entity (Transaction) with multiple specialized subtypes, and by abstracting shared information (such as Location) into reusable components, the design ensures flexibility, clarity, and ease of integration across various features in a digital wallet ecosystem.

In conclusion, the refined database design not only addresses the limitations of the original data structure but also provides a solid and extensible foundation for real-world digital payment systems like GOPAY—enabling secure operations, robust

transaction tracking, and long-term development by developers, analysts, and stakeholders alike.

C. SQL

	Table	Action	Rows	Type	Collation	Size	Overhead
☐	balance_topup_method	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	location	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	merchant_info	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	transaction_cashback	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	transaction_info	Browse Structure Search Insert Empty Drop	30	InnoDB	utf8mb4_general_ci	64.0 KiB	-
☐	transaction_payment	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	transaction_refund	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	transaction_topup	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	transaction_transfer	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	transaction_withdrawal	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	user_info	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	wallet_info	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
	12 tables	Sum	85	InnoDB	utf8mb4_general_ci	352.0 KiB	0 B

a) User Info

	location_id	street	city	province	postal_code	country
☐ Edit Copy Delete	LOC001	Jl. Sudirman No. 123	Jakarta	DKI Jakarta	10220	Indonesia
☐ Edit Copy Delete	LOC002	Jl. Gatot Subroto No. 456	Jakarta	DKI Jakarta	12930	Indonesia
☐ Edit Copy Delete	LOC003	Jl. Thamrin No. 789	Jakarta	DKI Jakarta	10350	Indonesia
☐ Edit Copy Delete	LOC004	Jl. HR Rasuna Said No. 101	Jakarta	DKI Jakarta	12940	Indonesia
☐ Edit Copy Delete	LOC005	Jl. Kuningan Raya No. 202	Jakarta	DKI Jakarta	12950	Indonesia

b) Wallet Info

	wallet_id	user_id	balance	coins_balance
Edit Copy Delete	WLT001	USR001	1500000	250
Edit Copy Delete	WLT002	USR002	2300000	180
Edit Copy Delete	WLT003	USR003	850000	420
Edit Copy Delete	WLT004	USR004	3200000	95
Edit Copy Delete	WLT005	USR005	1750000	310

c) Merchant Info

	merchant_id	merchant_name	acquirer_name	merchant_pan	terminal_id	qris_rrn	location_id
Edit Copy Delete	MERCH001	Warung Makan Sederhana	Bank ABC	1234567890123456	T001	923456789012	LOC001
Edit Copy Delete	MERCH002	Toko Elektronik Jaya	Bank XYZ	2345678901234567	T002	812345678901	LOC002
Edit Copy Delete	MERCH003	Apotek Sehat	Bank DEF	3456789012345678	T003	703456789123	LOC003
Edit Copy Delete	MERCH004	Minimarket 24	Bank GHI	4567890123456789	T004	691234567890	LOC004
Edit Copy Delete	MERCH005	Restoran Padang Minang	Bank JKL	5678901234567890	T005	845612347890	LOC005

d) Location

	location_id	street	city	province	postal_code	country
Edit Copy Delete	LOC001	Jl. Sudirman No. 123	Jakarta	DKI Jakarta	10220	Indonesia
Edit Copy Delete	LOC002	Jl. Gatot Subroto No. 456	Jakarta	DKI Jakarta	12930	Indonesia
Edit Copy Delete	LOC003	Jl. Thamrin No. 789	Jakarta	DKI Jakarta	10350	Indonesia
Edit Copy Delete	LOC004	Jl. HR Rasuna Said No. 101	Jakarta	DKI Jakarta	12940	Indonesia
Edit Copy Delete	LOC005	Jl. Kuningan Raya No. 202	Jakarta	DKI Jakarta	12950	Indonesia

e) Transaction Info

	transaction_id	user_id	from_wallet_id	to_wallet_id	transaction_type	amount	currency_code	status	created_at	time_hour
Edit Copy Delete	TRX001	USR001	NULL	WLT001	Top up	500000.00	IDR	Completed	2024-06-28	09:30:15
Edit Copy Delete	TRX002	USR001	WLT001	NULL	Withdrawal	200000.00	IDR	Completed	2024-06-28	14:30:15
Edit Copy Delete	TRX003	USR001	WLT001	NULL	Payment	75000.00	IDR	Completed	2024-06-28	12:30:15
Edit Copy Delete	TRX004	USR001	WLT001	WLT002	Transfer	150000.00	IDR	Completed	2024-06-28	16:30:15
Edit Copy Delete	TRX005	USR001	NULL	WLT001	Cashback	5000.00	IDR	Completed	2024-06-28	18:30:15
Edit Copy Delete	TRX006	USR001	NULL	WLT001	Refund	50000.00	IDR	Completed	2024-06-28	20:30:15
Edit Copy Delete	TRX007	USR002	NULL	WLT002	Top up	750000.00	IDR	Completed	2024-06-28	10:15:22
Edit Copy Delete	TRX008	USR002	WLT002	NULL	Withdrawal	150000.00	IDR	Completed	2024-06-28	15:15:22
Edit Copy Delete	TRX009	USR002	WLT002	NULL	Payment	125000.00	IDR	Completed	2024-06-28	13:15:22
Edit Copy Delete	TRX010	USR002	WLT002	WLT003	Transfer	200000.00	IDR	Completed	2024-06-28	17:15:22
Edit Copy Delete	TRX011	USR002	NULL	WLT002	Cashback	7500.00	IDR	Completed	2024-06-28	19:15:22
Edit Copy Delete	TRX012	USR002	NULL	WLT002	Refund	75000.00	IDR	Completed	2024-06-28	21:15:22
Edit Copy Delete	TRX013	USR003	NULL	WLT003	Top up	300000.00	IDR	Processing	2024-06-29	08:45:33
Edit Copy Delete	TRX014	USR003	WLT003	NULL	Withdrawal	100000.00	IDR	Failed	2024-06-29	09:45:33
Edit Copy Delete	TRX015	USR003	WLT003	NULL	Payment	95000.00	IDR	Completed	2024-06-29	10:45:33
Edit Copy Delete	TRX016	USR003	WLT003	WLT001	Transfer	120000.00	IDR	Processing	2024-06-29	11:45:33
Edit Copy Delete	TRX017	USR003	NULL	WLT003	Cashback	3000.00	IDR	Completed	2024-06-29	12:45:33
Edit Copy Delete	TRX018	USR003	NULL	WLT003	Refund	30000.00	IDR	Processing	2024-06-29	13:45:33
Edit Copy Delete	TRX019	USR004	NULL	WLT004	Top up	1000000.00	IDR	Completed	2024-06-29	11:20:45
Edit Copy Delete	TRX020	USR004	WLT004	NULL	Withdrawal	250000.00	IDR	Completed	2024-06-29	16:20:45
Edit Copy Delete	TRX021	USR004	WLT004	NULL	Payment	220000.00	IDR	Completed	2024-06-29	14:20:45
Edit Copy Delete	TRX022	USR004	WLT004	WLT005	Transfer	300000.00	IDR	Completed	2024-06-29	18:20:45
Edit Copy Delete	TRX023	USR004	NULL	WLT004	Cashback	10000.00	IDR	Completed	2024-06-29	20:20:45
Edit Copy Delete	TRX024	USR004	NULL	WLT004	Refund	100000.00	IDR	Completed	2024-06-29	22:20:45
Edit Copy Delete	TRX025	USR005	NULL	WLT005	Top up	600000.00	IDR	Completed	2024-06-29	13:55:12
Edit Copy Delete	TRX026	USR005	WLT005	NULL	Withdrawal	180000.00	IDR	Processing	2024-06-29	17:55:12
Edit Copy Delete	TRX027	USR005	WLT005	NULL	Payment	160000.00	IDR	Completed	2024-06-29	15:55:12
Edit Copy Delete	TRX028	USR005	WLT005	WLT001	Transfer	180000.00	IDR	Completed	2024-06-29	19:55:12
Edit Copy Delete	TRX029	USR005	NULL	WLT005	Cashback	6000.00	IDR	Completed	2024-06-29	21:55:12
Edit Copy Delete	TRX030	USR005	NULL	WLT005	Refund	60000.00	IDR	Completed	2024-06-29	23:55:12

f) Balance Top Up Method

	user_id	payment_method
Edit Copy Delete	USR001	Debit Card
Edit Copy Delete	USR002	Merchant
Edit Copy Delete	USR003	Merchant
Edit Copy Delete	USR004	Transfer
Edit Copy Delete	USR005	Transfer

g) Transaction Cashback

	transaction_id	cashback_source
Edit Copy Delete	TRX005	Merchant promo
Edit Copy Delete	TRX011	Loyalty program
Edit Copy Delete	TRX017	Seasonal bonus
Edit Copy Delete	TRX023	Referral bonus
Edit Copy Delete	TRX029	Welcome bonus

h) Transaction Payment

	transaction_id	merchant_id	order_id
Edit Copy Delete	TRX003	MERCH001	ORD001
Edit Copy Delete	TRX009	MERCH002	ORD002
Edit Copy Delete	TRX015	MERCH003	ORD003
Edit Copy Delete	TRX021	MERCH004	ORD004
Edit Copy Delete	TRX027	MERCH005	ORD005

i) Transaction Refund

	transaction_id	original_transaction_id
Edit Copy Delete	TRX006	TRX003
Edit Copy Delete	TRX012	TRX009
Edit Copy Delete	TRX018	TRX015
Edit Copy Delete	TRX024	TRX021
Edit Copy Delete	TRX030	TRX027

j) Transaction Top Up

transaction_id	payment_method
TRX001	Debit Card
TRX007	Merchant
TRX013	Merchant
TRX019	Transfer
TRX025	Transfer

k) Transaction Transfer

		transaction_id	receiver_user_id
☐	Edit Copy Delete	TRX016	USR001
☐	Edit Copy Delete	TRX028	USR001
☐	Edit Copy Delete	TRX004	USR002
☐	Edit Copy Delete	TRX010	USR003
☐	Edit Copy Delete	TRX022	USR005

l) Transaction Withdrawal

		transaction_id	withdrawal_bank
☐	Edit Copy Delete	TRX002	Bank Mandiri
☐	Edit Copy Delete	TRX008	Bank BCA
☐	Edit Copy Delete	TRX014	Bank BRI
☐	Edit Copy Delete	TRX020	Bank BNI
☐	Edit Copy Delete	TRX026	Bank CIMB